ELEC0447 – Project 2

Transmission System Steady-State Modeling,
Security, and Energy Transition Analysis

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1 Introduction and Objectives

The objective of this assignment is to deepen your understanding of modern transmission system analysis, operation, and grid planning using the open-source Python library PandaPower. Refer to the documentation of PandaPower for more information.

This project explores real-world challenges encountered by transmission system operators in the context of the energy transition, such as steady-state power flow and voltage/reactive power management (P-V curves, shunt compensation), primary frequency control and distributed slack power flow modeling, short-circuit current calculations and grid strength / rotor angle stability assessment, time-varying renewable generation dynamics (solar PV, wind, and demand duck curve), grid expansion and reinforcement, and $N-1$ security under massive load growth and remote renewable integration.

1.1 Benchmark: The IEEE 24-Bus Reliability Test System

We consider a standard, publicly available transmission grid benchmark: the **IEEE 24-Bus Reliability Test System (RTS)**. It has two voltage levels: a northern 230 kV extra-high-voltage (EHV) network (Buses 11–24) and a southern 138 kV high-voltage (HV) network (Buses 1–10). 5 transformers connect the 230 kV northern backbone to the 138 kV southern subtransmission grid. The northern area is rich in generation (hydro, nuclear, large thermal units, and vast renewable potential), whereas the southern area represents a dense, heavily loaded industrial and urban pocket.

Listing 1: The network can be loaded directly from PandaPower.

```
import pandapower as pp
import pandapower.networks as nw
net = nw.case24_ieee_rts()
```

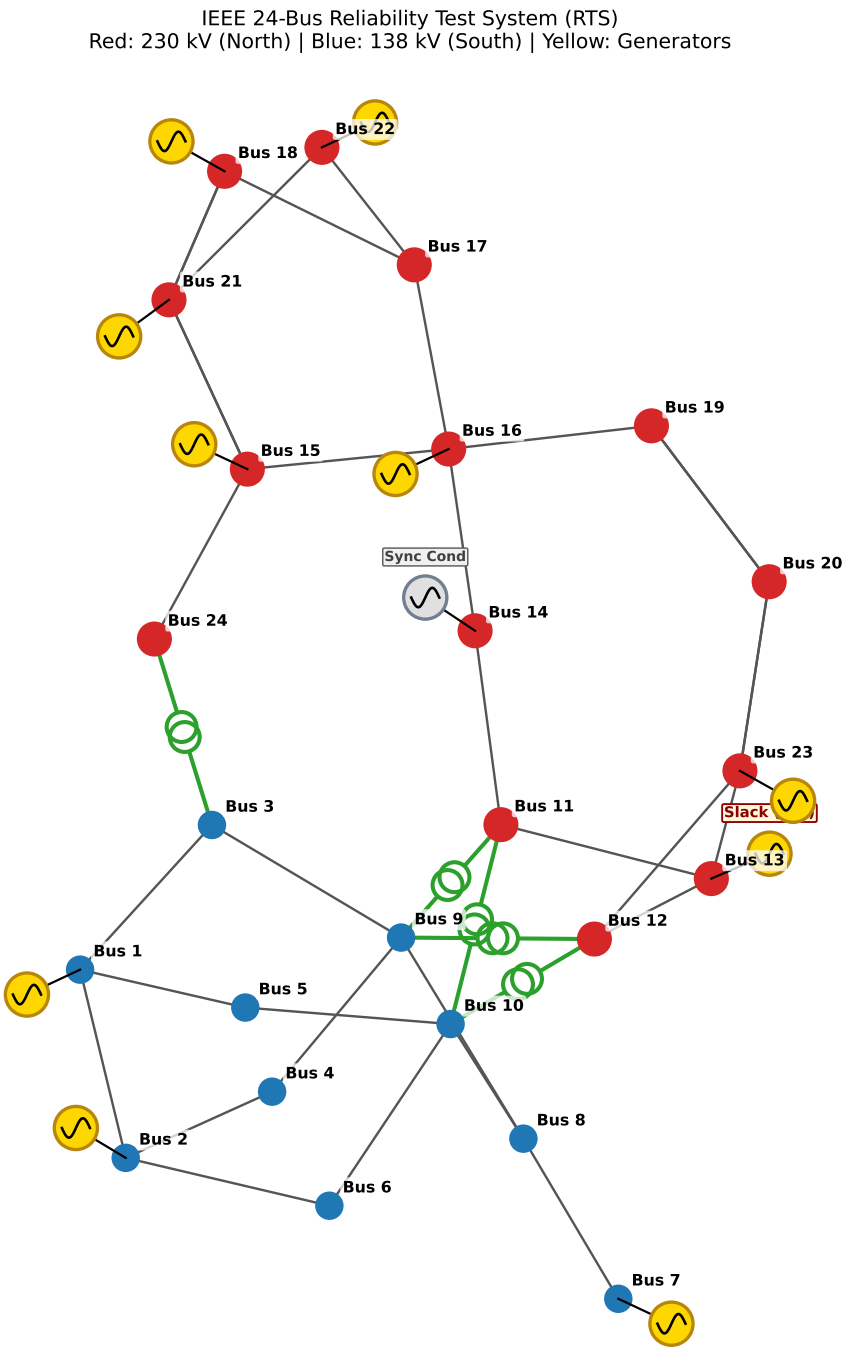


Figure 1: Geographical layout (North: 230 kV, South: 138 kV) with highlighted generation units and slack bus.

2 Questions

Unless explicitly stated, each question below must be solved independently, starting from the baseline grid. Include clear, interpretable, and properly formatted plots. Ensure that every graph supports your technical discussion.

2.1 Baseline AC Power Flow and Operating Envelope

Course References: *Lecture: Introduction to Power Flow Analysis (AC Newton-Raphson, bus types, DC power flow) and Lecture: Synchronous Generators (generator reactive limits $Q_{\min}/Q_{\max}$).*

- Run an AC Newton-Raphson power flow on the initial grid with generator reactive power limits enforced (`enforce_q_lims=True`).
- Display the voltage profile and branch loadings using matplotlib.
- Include an interactive plot of the network generated with Plotly (`pf_res_plotly`). Ensure the substation voltages and line loadings are clearly displayed, and that the **slack bus** (connected to the external grid reference) is highlighted.
- Are all substation voltages within standard statutory limits $[0.95, 1.05]$ pu?
- Which corridors or transformers are heavily loaded ($> 80\%$ of thermal capacity)?
- Run a DC power flow (`pp.rundcpp(net)`) and compare total transmission losses and branch flows with the AC results. Why are reactive power flows and voltage drops significant in EHV transmission?

2.2 Primary Frequency Response and Distributed Slack Power Flow

Course References: *Lecture: Frequency Control (governor droop σ, multi-generator primary power sharing, and analytical frequency deviation Δf).*

- Simulate the sudden loss (tripping) of the generating unit connected at Bus 23 ($P_{\text{lost}} = 155$ MW).
- **Single Slack Case**: Run the power flow using the default single slack bus. Observe the resulting generation redispatch and line loadings.
- **Distributed Slack Case**: Assuming all remaining units have a governor droop of $\sigma = 4\%$ (0.04), assign `slack_weight = sn_mva / droop` to each generator and re-run with `pp.runpp(net, distributed_slack=True)`.
- Compare the active power output adjustments (ΔP_i) of the generators and the loading of the transmission lines between single slack and distributed slack.
- Using the analytical formula from the course, calculate the expected quasi-steady-state frequency deviation Δf (assuming $f_N = 50$ Hz and damping $D \approx 0$).

For all subsequent steady-state power flow analyses (Question 3 onward), retain the distributed slack configuration (`distributed_slack=True`) so that power flow redispatch realistically reflects governor-droop sharing across the generator fleet.

2.3 Voltage Stability, P-V Nose Curves, and Shunt Compensation

Course References: *Lecture: Voltage Regulation and Voltage Instability (maximum power transfer, P-V nose curves, saddle-node bifurcation, generator reactive limits, and shunt compensation).*

- Disconnect the synchronous generator at Bus 1 and progressively ramp up the active power consumption of the load at Bus 3, maintaining a constant power factor.
- Plot the bus voltage magnitude as a function of the consumed active power (P-V curve) until the power flow diverges (why does it diverge?).
- Compare the P-V curves for:
  1. Enforcing generator reactive power limits vs. neglecting them.
  2. Two power factor values: $\cos\phi = 1.0$ and $\cos\phi = 0.95$ lagging.
- Determine the maximum permissible power transfer (P_{max}) and the voltage stability margin.
- The TSO wants to install a 100 MVAR shunt capacitor bank to improve the voltage profile and push back the collapse point. Test placing it at Bus 3 versus Bus 9. Motivate your choice with simulations.

2.4 Subtransient Short-Circuit Current Computation, Grid Strength, and Rotor Angle Stability

Course References: *Lecture: Transient Stability (subtransient reactance X''_d, IEC 60909 short circuits, Equal Area Criterion, CCT, synchronizing torque K_s, and Short-Circuit Ratio SCR) and Lecture: Introduction to Stability and Control Problems.*

- Configure the generator subtransient parameters ($X''_d = 0.20$ pu, $R_s = 0.05\,\Omega$, $\cos\phi = 0.85$) and external grid short-circuit level ($S''_{sc} = 5\,000$ MVA, $R/X = 0.1$).
- Run an IEC 60909 three-phase short-circuit calculation using `pandapower.shortcircuit.calc_sc(net, fault='3ph', case='max')`.
- Plot the initial symmetrical short-circuit current I''_k [kA] and short-circuit power S''_{sc} [MVA] across all 24 buses.
- Compare the northern 230 kV area with the southern 138 kV area. Which bus is electrically the *strongest*? Which is the *weakest*?
- **Link with Rotor Angle Stability**: Drawing upon the Equal Area Criterion and swing equations from the lectures:
  1. How does the local short-circuit power S''_{sc} (and Thévenin reactance X_{th}) influence the peak electrical power P_{max}, the decelerating area (Area B), and the Critical Clearing Time (CCT) of a synchronous generator following a three-phase fault?
  2. How does a strong grid (X_{th} small) affect the small-signal synchronizing torque coefficient K_s and the spatial spread of fault-induced voltage dips?
- Suppose a developer plans to connect a 500 MW offshore wind farm. Evaluate the Short-Circuit Ratio (SCR) if connected to Bus 3 versus Bus 15. What are the operational implications for grid strength and stability?

2.5 Time-Varying Data and Renewable Generation Variability

Course References: *Lecture: Introduction to Stability and Control Problems (net load variations, Duck curve phenomenon) and Lecture: Voltage Regulation (impact of RES on grid operating envelope).*

- Consider the 24-hour normalized profiles for system demand, solar PV generation, and wind power generation provided in `profiles_24h.csv` on eCampus.
- To reflect day-ahead operational planning, the dataset also includes a pre-scheduled conventional generation production plan (`conv_dispatch_factor`), balancing expected net demand ($D - P_{\text{solar}} - P_{\text{wind}}$), along with technology breakdowns (hydro, nuclear, coal, gas).
- Equip the network with 500 MW of solar PV in the south (250 MW each at Bus 3 and Bus 5) and 250 MW of wind in the north (Bus 19).
- For each hour $h \in \{0, \ldots, 23\}$, update system active and reactive loads, set renewable generation, scale conventional active setpoints with `conv_dispatch_factor`, and run the power flow with distributed slack enabled (`distributed_slack=True , enforce_q_lims=True`).
- Plot the daily trajectory of:
  1. Total system active power demand, solar, wind, and conventional dispatch (stacked area or lines).
  2. The maximum transmission line loading percentage over time.
  3. Minimum and maximum substation voltages over the 24 hours.
- At which hour is the transmission network under the greatest thermal stress? At which hour do we observe the lowest voltage?
- Discuss the "duck curve" phenomenon: how does midday solar PV relieve transmission loading and fuel consumption, and why does the sudden loss of solar generation at dusk combined with peak demand create severe ramping and reactive power / voltage challenges?

2.6 Grid Expansion: Massive Load Growth, Remote Renewable Boom, and $N-0$ Reinforcements

Course References: *Lecture: High Voltage DC Transmission (VSC-HVDC operating principles, four-quadrant control, and power flow modeling with `pp.create_dcline`) and Lecture: Transformers (Phase-Shifting Transformers).*

- To reflect the decarbonization of industry and heating, the active and reactive demand in the southern region (Buses 1–10) increases permanently by $+40\%$.
- In parallel, $1\,500$ MW of fossil-fired generation in the south is decommissioned, replaced by $2\,500$ MVA of remote wind generation connected at the northern extremity (Bus 22), injecting $1\,500$ MW under high-wind conditions.
- Simulate the intact system ($N-0$ state):

  - Which transmission corridors connecting the north to the south are severely overloaded ($> 100\%$)? How is the bus voltage profile impacted across the 138 kV and 230 kV subsystems?
  - How does the decommissioning of southern synchronous machines affect the short-circuit level (S''_{sc}) and grid strength in the south? Based on the lecture discussion, what are the direct consequences for the rotor angle stability of remaining synchronous units?

- **Initial $N-0$ Grid Reinforcement**: Propose and implement modifications to the grid to restore acceptable $N-0$ steady-state operation (all line loadings $\leq 100\%$ and substation voltages within $[0.95, 1.05]$ pu). You may consider AC solutions (line duplication, Phase-Shifting Transformers) or an embedded VSC-HVDC link (`pp.create_dcline`). Validate your proposed design.

2.7 $N-1$ Security Screening and Additional Reinforcements for $N-1$ Compliance

Course References: *Lecture: High Voltage DC Transmission (active flow redirection and controllable corridors) and Lecture: Frequency Control (system security assessment & $N-1$ criteria).*

- Starting directly from your $N-0$ reinforced grid from Section 2.6, perform an automated $N-1$ contingency screening by taking out one transmission branch at a time.
- Does your $N-0$ design satisfy the $N-1$ security criterion? Identify the critical single-line contingencies that still trigger post-fault line overloads ($> 100\%$) or voltage instability.
- **Additional Reinforcements for $N-1$ Compliance**: Propose and implement additional or alternative reinforcements (e.g., targeted corridor duplications on remaining bottleneck lines, fine-tuning HVDC setpoints, or Phase-Shifting Transformers) to make the network fully $N-1$ compliant.
- Quantitatively compare the three grid states: (i) Unreinforced Grid, (ii) $N-0$ Reinforced Grid, and (iii) Full $N-1$ Compliant Grid in terms of maximum loading, contingency overloads ($> 100\%$ and $> 120\%$), total active power transmission losses, and bus voltage profiles.

3 Submission and Evaluation Rules

1. **Group work**: Work in groups of 2 students, but **defense is individual** during the oral examination.
2. **Deliverables on eCampus for December 18, 2026 at 23:59.**:
   - A clean, well-commented Jupyter notebook (`.ipynb`) allowing full reproducibility of all results and figures, and including a discussion of the results.
   - All plots must have labeled axes with engineering units, clear legends, and readable fonts.